PRAGNA MAMIDIPAKA

Curriculum Vitae

Ph.D. Candidate, Carnegie Mellon University
Department of Electrical and Computer Engineering
Pittsburgh, PA, USA

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Personal Website

BRIEF BIOGRAPHY

My research focuses broadly on observability, making complex systems easier to understand, diagnose, and improve. My approach has been to develop abstractions that summarize system execution, making system behavior easier to reason about. I have applied this perspective to data center congestion control, where a control-path abstraction allows us to model algorithm execution, and subsequently guide workload selection. In another work, we check for unintended behaviors arising from interactions among eBPF network functions, by extracting and organizing dependencies from bytecode. Looking ahead, I am interested in autonomous observability and finding the right system support for agent-driven management. My research has been recognized with the 2026 Prabhu and Poonam Goel Graduate Fellowship and the 2024 Carnegie Institute of Technology Dean's Fellowship.

EDUCATION

Ph.D., Electrical and Computer Engineering 2024–2029 (expected)
Carnegie Mellon University, Pittsburgh, PA *Grade: 4.00/4*
Advisor: Prof. Theophilus Benson

B.Tech., Electrical Engineering 2020–2024
Indian Institute of Technology Hyderabad, India *Grade: 9.75/10*
Minor in Artificial Intelligence
President of India Gold Medallist

High School and Senior Secondary Education 2007–2020
City International School, Pune, India; Science discipline
Ranked first in the school in the CBSE board examination.

HONORS AND AWARDS

Prabhu and Poonam Goel Graduate Fellowship 2026
Awarded by the Carnegie Mellon University ECE Department.

Carnegie Institute of Technology Dean's Fellow 2024–2025
Full tuition support and stipend for one year of Ph.D. study.

President of India Gold Medal 2024
Highest overall CGPA among all graduating B.Tech. students at IIT Hyderabad.

Institute Silver Medal 2024
Highest CGPA in the Electrical Engineering Department at IIT Hyderabad.

Academic Excellence Award 2024; 2022
Recognized for outstanding academic performance.

Excellence in Innovation Award 2022
Among the top 5 of 1,300 participants in the Indo-Japanese Road2Shine program.

All India Rank 3418, JEE-Advanced 2020

CBSE Merit Certificate 2018
Top 0.1% in Grade 10 Science among 1.7 million candidates.

PUBLICATIONS AND PREPRINTS

Systems and Networking

- Pragna Mamidipaka, Srikanth Sundaresan, Theophilus A. Benson.
  Title omitted to maintain anonymity.
  Under submission to ACM SIGMETRICS 2027.
- Animesh Singh, K Shiv Kumar, S. VenkataKeerthy, Pragna Mamidipaka, R V B R N Aaseesh, Sayandeep Sen, Palanivel Kodeswaran, Theophilus A. Benson, Ramakrishna Upadrasta, Praveen Tammana.
  Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behavior.
  ACM CoNEXT, 2026.
- Pragna Mamidipaka, Srikanth Sundaresan, Theophilus A. Benson.
  Towards Truly Burst-Aware Evaluation of Data Center Congestion Control.
  Asia-Pacific Workshop on Networking (APNet), 2026.
- Pragna Mamidipaka, Theophilus A. Benson.
  Apiary: Distributed Programming and Lifecycle Management for eBPF.
  N2Women Workshop at SIGCOMM, 2025.

Pulsar Astronomy

- The Indian Pulsar Timing Array Data Release 2: I. Dataset and Timing Analysis.
  Publications of the Astronomical Society of Australia, 2025.
- Low-frequency pulse-jitter measurement with the uGMRT I: PSR J0437–4715.
  Publications of the Astronomical Society of Australia, 2024.
- Application of Efron-Petrosian method to radio pulsar fluxes.
  Journal of Cosmology and Astroparticle Physics, 2024.
- Do Pulsar and Fast Radio Burst dispersion measures obey Benford's law?
  Astroparticle Physics, 2023.

PROFESSIONAL EXPERIENCE

Software Developer, Arcesium May 2023–Jul 2023
Hyderabad, India
Developed a CI/CD workflow analysis application with dashboards in Apache Superset and Grafana.

Undergraduate Researcher, University of Tokyo Dec 2022
Tokyo, Japan; advisor: Prof. Kunihiko Sadakane
Studied succinct encoding of ordinal trees using tree covering during a research internship.

Student Member, Indian Pulsar Timing Array Jun 2022–Jun 2024
Hyderabad, India
Contributed to the InPTA Consortium's Data Reduction team. Worked with the National Centre for Radio Astrophysics to collate legacy data from the upgraded Giant Metrewave Radio Telescope (uGMRT).

5G Testbed Intern, Wisig Networks Jan 2022–Jun 2022
Hyderabad, India
Developed a module to transmit Downlink Control Information (DCI) from a gNB base station to user equipment, gaining experience with L2/L3 protocol stack development.

PROFESSIONAL SERVICE

Artifact Evaluation Committee Member, CAIS 2026
Served on the Artifact Evaluation Committee for the first ACM Conference on AI and Agentic Systems.

Shadow Program Committee Member, IMC 2026

Shadow Program Committee Member, ACM CoNEXT 2025

Evaluator, Meeting of the Minds, Carnegie Mellon University 2025
Reviewed senior undergraduate research projects at CMU's annual research symposium.

TEACHING AND COMMUNITY OUTREACH

K–12 Community Outreach 2026
Focus on making engineering more accessible and equitable in K–12 spaces and re-envisioning engineering education pedagogy.

Yoga Instructor, Carnegie Mellon University 2024–Present
Teach weekly group yoga sessions through CMU's GroupX fitness program.

Volunteer Teacher, National Service Scheme, India 2022–2024
Taught Grade 10 mathematics at rural schools and led motivational lectures as part of the Aksharamaala program.